

```
#include <TFT_eSPI.h>

TFT_eSPI tft = TFT_eSPI();

const int buttonPin = 0; // Button pin for jumping (use INPUT_PULLUP)

int xP = 320; // Pillar x position
int yP = 100; // Pillar gap y position (bottom of upper pillar)
int yB = 50; // Bird y position
int movingRate = 3; // Pillar movement speed
float fallRate = 0;
int fallRateInt = 0;
int score = 0;
int lastSpeedUpScore = 0;
int highestScore = 0; // Simple variable, no EEPROM
boolean gameStarted = false;
boolean buttonPressed = false; // For debounce

// Colors in RGB565 format
uint16_t skyColor = tft.color565(114, 198, 206);
uint16_t groundColor = tft.color565(221, 216, 148);
uint16_t grassColor = tft.color565(47, 175, 68);
uint16_t pillarColor = tft.color565(0, 175, 0);
uint16_t pillarBorderColor = TFT_BLACK;
uint16_t birdColor = TFT_YELLOW;
uint16_t textColor = TFT_BLACK;

void setup() {
  pinMode(buttonPin, INPUT_PULLUP);
  randomSeed(analogRead(0)); // For random pillar positions

  tft.init();
  tft.setRotation(1); // Landscape mode: 320 width x 240 height
  tft.fillRect(skyColor);
```

Fig. 9. Code snippet of a game developed for the device



Fig. 10: Game console running a game

code in this article.

Setup steps:

1. Install the TFT_eSPI library and the ESP8266 board package in the Arduino IDE

2. Configure User_Setup.h for the chosen display (see [TFT_eSPI documentation](https://doc-tft-espi.readthedocs.io/starting))

3. Place the board in download mode by connecting GPIO0 to GND. Connect RX, TX, and power via the USB solder pads to an FTDI adaptor, then upload the sketch

4. Optionally, enable OTA updates to upload games over Wi-Fi

Testing

After uploading the code and assembling the device, connect the

battery and test the games. The device can also be repurposed with Wi-Fi Deauther firmware or other Wi-Fi tools for experimentation. Users must observe applicable laws and ethical guidelines. Fig. 10 shows the console running a game. **EFY**

Bonus. You can watch the video of the tutorial of this DIY project at <https://www.electronicsforu.com/videos-slideshows/live-diy-designing-smallest-gameboy>

TO DOWNLOAD SOURCE CODE AND RESOURCES



Scan this QR code or go to <https://sourcecode.electronicsforu.com> to access and download source code and other resources for this project

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-Jeff Bezos

